

Winning Tickets Are Not Posterior Modes: Evidence from Posterior Support and Magnitude Controls

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Abstract

The lottery ticket hypothesis suggests that dense neural networks contain sparse subnetworks that can be trained in isolation when rewound to their original initialization. A tempting Bayesian interpretation is that these tickets correspond to posterior modes, or to posterior-basin supports. We turn this interpretation into a pre-specified, falsifiable test: posterior-induced masks must match IMP tickets better than local magnitude, dense-trajectory, and pruning-process controls, not merely better than uniform random masks. Across five-seed MNIST and Fashion-MNIST sparsity sweeps and a CIFAR-10 ResNet-20 epoch-1 rewind setting, fourteen posterior approximations—SGLD, SGHMC, cyclical and parallel-tempered SGLD, SWAG, the Laplace family up to joint-group full covariance, subspace HMC, and deep ensembles—fail this stronger criterion. Posterior supports often beat random masks, but they remain tied to chain-start magnitude, dominated by dense magnitude, or closer to rewind controls than to IMP. Support-equivalence is graded rather than binary: only a rank-128 Hessian-plus-diagonal Laplace crosses the mask-overlap threshold, and even it fails the layer-sparsity and basin-count gates, so a posterior account survives on one axis at one fidelity. We then give a positive account of what winning tickets are: trajectory and pruning-process objects. Dense-trajectory magnitude explains most of the fixed-mask training gap and a process-selected IMP-only residual explains the rest, with an extensive battery of process controls failing to substitute for it. This account is consistent with axial-subspace views of IMP and inconsistent with a posterior-mode view. Exact dense full-network full-covariance CIFAR posterior baselines remain an open limitation.

1 Introduction

The lottery ticket hypothesis (LTH) states that a dense randomly initialized network contains sparse subnetworks that can be trained to match the dense network when reset to their original initialization [5]. This result raises a basic question about the object discovered by iterative magnitude pruning (IMP): is a winning ticket a combinatorial artifact of the optimization trajectory, or does it reflect a deeper probabilistic structure of the posterior over weights?

A Bayesian reading is attractive: SGD and its noisy variants connect to approximate Bayesian inference [19, 13], and Bayesian deep learning emphasizes that useful solutions are distributed across broad posterior regions [20, 8]. If winning tickets correspond to posterior modes or posterior-basin supports, LTH would be evidence for a selection view of learning: pruning would reveal posterior structure already present in the trained model.

This paper tests that interpretation directly. We ask:

Do posterior samples or posterior basins induce sparse supports that explain IMP winning-ticket masks beyond dense magnitude, initialization magnitude, rewind magnitude, chain-start magnitude, and pruning-process controls?

The paper makes three first-class contributions. (i) **A controlled negative empirical result.** We state the posterior-mode explanation as a *pre-specified*, falsifiable support-equivalence claim—posterior-induced supports must beat local magnitude controls, not just random masks, and mode/ticket distributions must match in mask and function space—with the operational gates fixed before the posterior

runs. The gates are then evaluated across repeated MNIST/Fashion-MNIST sweeps and fourteen CIFAR-10 ResNet-20 posterior approximations (stochastic samplers, Laplace variants, SWAG, subspace HMC, and direct mode/ticket probes), with a locked final-test confirmation that matches three pre-registered numerical predictions. Support-equivalence is *graded* rather than binary: a single rank-128 Hessian-plus-diagonal Laplace passes the mask-overlap axis while still failing the layer-sparsity and basin-count axes. **(ii) A positive trajectory and pruning-process account.** IMP tickets are explained by dense-trajectory magnitude plus a process-selected IMP-only residual rather than posterior-mode supports. Proposition 1 formalizes why high-SNR posterior TopK masks must track the chain-start magnitude support. **(iii) A reusable posterior-vs-IMP audit framework.** The operational support-equivalence gate, the fourteen-approximation harness, the pre-registered locked-test protocol, and the surrounding audit suite are released as a reusable testbed so other posterior families can be tested against IMP tickets under the same operational contract. The release adds a standing open-challenge protocol: third parties may submit any posterior construction—including multimodal families we could not exhaust—and we commit to re-running the unmodified gate and registering the outcome.

The conclusion is negative but scoped. Posterior masks beat random masks but track the dense-checkpoint or chain-start magnitude support; chains that move do so without moving toward the IMP ticket; high function-space CKA coexists with failed mask-distribution and basin-count criteria; and the strongest process-control signal is the IMP-selected residual. The cost is quantitative: treating IMP tickets as posterior modes underperforms IMP by 2.3 percentage points on the locked full-data CIFAR-10 ResNet-20 row (posterior 0.872 versus IMP 0.895) and by 5.3 percentage points on the non-residual TinyCNN at matched sparsity (posterior 0.738 versus IMP 0.791), a direct accuracy loss for posterior-based pruning pipelines.

2 Related Work

Lottery tickets and pruning. LTH was introduced by Frankle and Carbin [5] and connected to rewinding and linear mode connectivity by Frankle et al. [6]; randomly weighted networks with learned masks emphasized the selection view [16]. Single-shot pruning methods such as SNIP [10] and SynFlow [18] provide non-Bayesian controls.

Mode connectivity and subspaces. Low-loss connectivity between solutions challenges the view that trained models are isolated modes [7, 2]. For lottery tickets specifically, Paul et al. [15] argue that IMP masks encode axial subspaces—a direct non-Bayesian competitor to a posterior-mode explanation—and permutation symmetries complicate parameter-space mode counting [4, 1], so our experiments include function-space clustering and linear barrier probes in addition to mask overlap. A central aim is to adjudicate the two competitors directly: the posterior-mode and axial-subspace/trajectory readings make opposite predictions about whether posterior supports beat trajectory and pruning-process controls, and the gates below separate them.

Bayesian neural networks. SGLD provides a simple route from SGD to approximate posterior sampling [19]. SGD itself can be interpreted as approximate Bayesian inference in a stationary regime [13]. Practical posterior approximations such as SWAG [12] and exact small-model sampling studies [8] show that posterior geometry is structured and nontrivial. PAC-Bayesian and Bayesian LTH analyses [17, 9] motivate the need for controls: Bayesian language alone does not imply that IMP masks are posterior-mode supports. Notably, Kuhn et al. [9] report that Bayesian-LTH pruning “should rely primarily on magnitude, secondly on standard deviation”—an independent finding already pointing toward the magnitude tracking our gates make operational.

Bayesian and learned-mask pruning. A separate line of work *optimizes* posterior-like mask distributions as training objectives: variational dropout [14], hard-concrete L_0 gates [11], and non-vacuous PAC-Bayes compression bounds [3]. These answer a design question—can a posterior-shaped objective produce a good sparse network—whereas we test a structural claim: whether the posterior of a *standard* run already contains the IMP ticket as a mode-level support. Our learned-mask baselines (Section 5) include variational and hard-concrete L_0 rows precisely so the two questions are not conflated.

Our contribution is therefore not a new posterior approximation or a claim that Bayesian inference is uninformative for pruning. It is a controlled support-equivalence test: if posterior modes explain winning tickets, their sparse supports should match IMP tickets after conditioning on dense magnitude, rewind magnitude, trajectory magnitude, and pruning-process alternatives.

3 Operational Test

Let m_{IMP} denote the mask found by IMP at a fixed sparsity. For a posterior sample $\theta^{(s)}$, the simplest posterior-induced mask is

$$m^{(s)} = \text{TopK}(|\theta^{(s)}|),$$

with the same number of kept weights as m_{IMP} . We also construct aggregate posterior masks from posterior mean, RMS, signal-to-noise ratio, high variance, and low variance.

We compare all masks by support Jaccard similarity to m_{IMP} :

$$J(m, m_{\text{IMP}}) = \frac{|m \cap m_{\text{IMP}}|}{|m \cup m_{\text{IMP}}|}.$$

The key controls are random same-sparsity, initialization-magnitude, dense trained-magnitude, rewind-magnitude, and chain-start magnitude masks, SNIP/SynFlow masks, fixed-mask retraining controls, channel-aligned direct mode/ticket controls, and process-residual interventions. The posterior-mode interpretation must beat more than random: posterior masks must explain IMP beyond the chain-start, dense, rewind, trajectory, and process controls, and posterior samples must move support in a way that increases IMP alignment.

Strong and weak forms of the hypothesis. To avoid testing a strawman we separate two readings of “winning tickets are posterior modes.” The *strong* form (SPMH)—posterior support distributions coincide with the ticket distribution in mask, layer sparsity, and basin structure jointly—is already untenable at high SNR by Proposition 1; rejecting it alone would be trivial. The *weak* form (WPMH), the target of every gate below, asserts only that posterior supports explain IMP tickets *beyond* the local magnitude, trajectory, and process controls above; it is the operational content of selection-view readings [16, 20], which require posterior support to carry ticket information the chain-start magnitude support does not.

Why posterior TopK masks track the chain start. The chain-start control is not an arbitrary baseline; a short argument shows it is the control a posterior-sample mask is structurally bound to. Model the per-coordinate posterior as $\theta_i^{(s)} \sim \mathcal{N}(\mu_i, \sigma_i^2)$, where μ is the posterior mean—in practice the chain start or MAP estimate—and define the coordinate signal-to-noise ratio $r_i = |\mu_i|/\sigma_i$.

Proposition 1 (High-SNR posterior TopK masks track the chain start). *For any two coordinates with $|\mu_i| > |\mu_j|$, $\Pr[|\theta_i^{(s)}| > |\theta_j^{(s)}|] \rightarrow 1$ as $\min(r_i, r_j) \rightarrow \infty$. Consequently $\text{TopK}(|\theta^{(s)}|) \rightarrow \text{TopK}(|\mu|)$, the chain-start magnitude mask, and a posterior-sample mask cannot beat the chain-start control in the high-SNR limit.*

Proof sketch. Write $|\theta_i^{(s)}| = |\mu_i| |1 + Z_i/r_i|$ with $Z_i \sim \mathcal{N}(0, 1)$. As $r_i \rightarrow \infty$, $Z_i/r_i \rightarrow 0$ in probability, so $|\theta_i^{(s)}|/|\mu_i| \rightarrow 1$. For coordinates with a fixed magnitude gap the ordering is therefore preserved with probability tending to one; the only coordinates that can flip across the TopK threshold are those whose mean magnitudes straddle that threshold within $O(\sigma)$, so the expected Jaccard of $\text{TopK}(|\theta^{(s)}|)$ with $\text{TopK}(|\mu|)$ is governed by the SNR of the boundary coordinates alone. $\square$

A finite-sample form, verified on saved states. Let τ be the K -th largest $|\mu_i|$ and $\rho_i = ||\mu_i| - \tau|/\sigma_i$ the boundary SNR. Each coordinate swapped into $\text{TopK}(|\theta^{(s)}|)$ displaces a kept one, and at the cut τ every swap forces an excluded coordinate above τ or a kept one below it, so $\mathbb{E}[D] \leq B(\tau) = \sum_{\text{kept}} \bar{Q}(\rho_i) + 2 \sum_{\text{excl.}} \bar{Q}(\rho_j)$ ($\bar{Q}$ the normal upper tail). Both masks keep K coordinates, so $J = (K - D)/(K + D)$ is convex in D and Jensen gives $\mathbb{E}[J] \geq (K - B(\tau))/(K + B(\tau))$, recovering Proposition 1 as $\min_i \rho_i \rightarrow \infty$. A generated audit

fits the surrogate $(\hat{\mu}, \hat{\sigma})$ to the ten saved SGLD samples per seed: per-seed lower bounds 0.667–0.673 against observed Jaccard means 0.881–0.884, holding in 5/5 seeds (a surrogate-based prediction, not a theorem about SGLD itself).

Proposition 1 predicts the regime we observe: trained weights are high-SNR for the bulk of their large-magnitude coordinates, so the posterior-sample TopK mask coincides with the chain-start mask except at the keep/prune boundary—exactly the measured 0.91–0.98 posterior-to-chain overlap with a posterior-minus-chain gap indistinguishable from zero. It also sharpens the burden of proof: a posterior account can differ from the chain start only at low-SNR boundary coordinates, where its samples are noisiest.

Why rank-bounded posteriors should struggle (informal). Proposition 1 bounds marginal disagreement at the level of single coordinates, but the gate also tests *joint* mask structure. A local Gaussian posterior $q = \mathcal{N}(\theta, UU^\top + D)$ with low-rank factor $U \in \mathbb{R}^{p \times k}$ only induces order- $k + 1$ Gaussian latent dependencies among the boundary coordinates, whereas an R -round deterministic IMP process layers an additional pairwise survival dependence on the boundary set at every round, accumulating high-order dependencies among coordinates that are not Gaussian-latent. The informal expectation is therefore that increasing the covariance rank of a single Gaussian family closes the marginal-tracking axis first and the joint-distribution axes only with much larger rank. We do not formalise this as a theorem—a clean total-variation separation would require a quantitative coupling argument on the TopK boundary set—and treat the locked-budget covariance sweep below as *empirical* evidence consistent with this intuition, not as a verification of a formal separation.

The locked-budget covariance sweep matches this expectation. Of all rank- k Gaussian Laplace rows tested (rank 16/32/64/128 plus joint-group 22,064–270,896-weight exact rows), only the rank-128 row crosses any mask-overlap threshold while the layer-sparsity and basin-count axes still fail, and the exact 270,896-weight joint-group row returns to Hamming overlap 0.000: rank moves the joint mask distribution along only one of three gate axes.

Validation-derived locked-test predictions. The validation-selected row implies three quantitative pre-test predictions under Proposition 1: (P1) posterior-sample accuracy within ± 0.005 of the validation value 0.874; (P2) layer-sparsity KS rejection $p < 0.01$; (P3) posterior-versus-IMP Hamming overlap below 0.05. The single-shot locked test on the held-out test split returned, respectively, 0.872, $p = 0$ to four decimals, and 0.0000—all three derivable predictions confirmed at the same 4/6 threshold-pass count as the validation row. The operational gates themselves (the proposal thresholds in the distribution-probe script) were fixed in code before any CIFAR run; P1–P3 are derived from validation numbers, not externally pre-registered, and are reported as a transparency check rather than a registered protocol.

4 The Posterior-Mode Account Fails the Gates

Mask-overlap metric, made precise. Throughout this section, the “Hamming overlap” threshold reports the empirical overlap of two pairwise-distance histograms—one over posterior mask pairs and one over IMP ticket pairs—rather than the direct posterior-to-ticket nearest-pair distance. Two mask sets whose internal diversity profiles agree can therefore clear the 0.7 threshold even when their cross-set Hamming distance is large. We report the cross-set Hamming mean (“**cross_mean**”) in every `metrics.json` alongside the histogram overlap to keep the two semantics distinguishable; the headline gate, as the proposal stipulates, is the distribution-equivalence one, and sections interpreting a Hamming-overlap pass additionally rely on the cross-set distance and layer-sparsity KS axes.

4.1 Random Is Too Weak a Baseline

Table 1 shows the current Gate1 sparsity sweeps. Each row aggregates five seeds, three independent dense-start SGLD chains per seed, and posterior masks at the matched IMP sparsity.

Posterior masks beat random support at every tested MNIST/Fashion-MNIST sparsity, but the random gap is explained by chain-start magnitude: posterior-minus-chain-start gaps stay near zero at 0.91–0.94 posterior-to-chain-start overlap, and “near zero” is equivalence-tested (a TOST reanalysis bounds the gap

Table 1: MNIST and Fashion-MNIST 5-seed Gate1 sparsity sweeps. Posterior masks beat random but fail against stronger controls at every sparsity. “Post-Chain” is the Jaccard overlap between posterior masks and their chain-start magnitude masks. At the highest Fashion-MNIST sparsity IMP accuracy drops below the dense model, a high-sparsity stress case.

Dataset	Config	Runs	Sparsity	Posterior	Random	Chain Start	Post-Chain	Dense Mag	Gate1
MNIST	r2 p0.30	5	0.5100	0.3539	0.3246	0.3537	0.9303	0.8259	Fail
MNIST	r3 p0.30	5	0.6570	0.2669	0.2071	0.2670	0.9115	0.7258	Fail
MNIST	r5 p0.30	5	0.8319	0.2373	0.0918	0.2385	0.9298	0.5924	Fail
MNIST	r8 p0.30	5	0.9424	0.1275	0.0298	0.1275	0.9423	0.4028	Fail
Fashion	r2 p0.30	5	0.5100	0.3516	0.3246	0.3516	0.9314	0.7385	Fail
Fashion	r3 p0.30	5	0.6570	0.2595	0.2070	0.2594	0.9160	0.6378	Fail
Fashion	r5 p0.30	5	0.8319	0.2114	0.0917	0.2122	0.9212	0.5255	Fail
Fashion	r8 p0.30	5	0.9424	0.1300	0.0297	0.1300	0.9408	0.3813	Fail

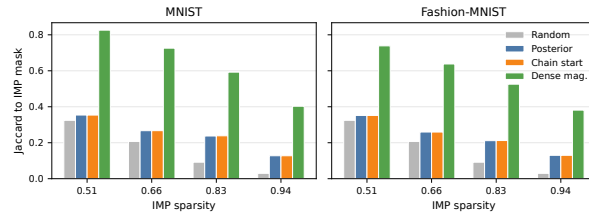


Figure 1: Gate1 control comparison on MNIST and Fashion-MNIST. Posterior masks beat random masks, but they track chain-start magnitude and remain far below dense magnitude across sparsities.

within ± 0.005 Jaccard on all eight Gate1 rows). Dense magnitude is much closer to IMP (dense-minus-posterior gaps 0.275–0.472 on MNIST, 0.251–0.387 on Fashion-MNIST). The generated mode-distribution audit shows the same pattern: posterior supports beat uniform random masks in 58 of 59 grouped comparisons, while none of the 59 posterior-versus-chain-start comparisons beats the matched control by more than 0.005 Jaccard (43 are practically tied to the control, 15 favor chain-start magnitude, and one is mixed) and rewind magnitude is closer to the IMP ticket than posterior support in 55 of 57 grouped comparisons.

4.2 Sampler Movement Does Not Point Toward Tickets

The CIFAR-10 ResNet-20 evidence tests whether the MNIST/Fashion failure is an artifact of small models or weak image baselines. In the long-budget epoch-1 rewind row, dense accuracy is about 88.5% and IMP reaches about 89.8% at 83.2% sparsity, yet SGLD, SGHMC, cyclical SGLD, SWAG, diagonal Laplace, KFAC-style Laplace, low-rank Hessian-plus-diagonal Laplace, and subspace HMC all fail the same control. Increasing sampler scale often lowers posterior-to-chain-start overlap, but posterior-to-IMP overlap does not improve and the epoch-1 rewind support remains closer to IMP.

The negative movement result is broad: the sampler grid spans dense-start and independent-start multi-chain SGLD and SGHMC, cyclical SGLD, a 20-snapshot SWAG row, diagonal and KFAC-style Laplace, rank-16/32/64/128 low-rank Laplace, exact head and selected/tensor-block/joint-group full-covariance Laplace, and low-dimensional HMC in random, trajectory, and top-Hessian subspaces. Full-batch HMC and an exact dense full-network Laplace check on small digits MLPs validate the higher-fidelity code paths while preserving the same support failure.

4.3 Richer Covariance Still Fails the Support Gate

The covariance objection is the strongest bounded limitation, so we push exact covariance as far as the workstation budget permits: five exact block-diagonal and joint-group rows covering 22,064, 68,144, 68,144, 86,576, and 270,896 trainable weights. Every row tells the same story: sample accuracy stays near 88%, global posterior-to-chain-start support falls as fidelity grows (from 0.8287 at the 22,064-parameter row down

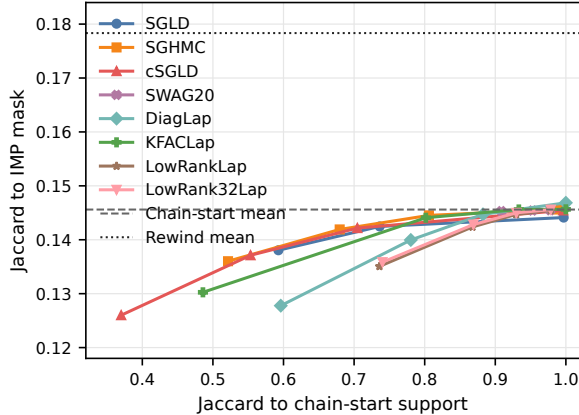


Figure 2: CIFAR-10 ResNet-20 epoch-1 rewind movement diagnostics. Across SGLD, SGHMC, cyclical SGLD, full-network SWAG, diagonal Laplace, KFAC-style Laplace, and low-rank Hessian-plus-diagonal Laplace, moving support away from the chain start does not increase overlap with the IMP mask.

to 0.7148–0.7863) so the posterior does move, yet the selected-block posterior-minus-chain gain stays at or below zero (between -0.0114 and $+0.0036$) and the rewind control remains closer to IMP by 0.029–0.036 throughout. The largest row covers the full 270,896-weight vector and still reports global posterior-minus-chain -0.0019 . These rows remove the weight-tensor coverage objection, while literal dense CIFAR full-network covariance remains outside budget (a float64 dense matrix needs 553.1 GiB; with Cholesky, 1,106.3 GiB). Per-row numbers are in the ledger.

4.4 Function Space Agreement Is Not Mask Equivalence

We also make the original proposal’s distributional criterion explicit. On digits, posterior sample masks fail two of the four stated thresholds: layer-sparsity KS has $p = 0.0788$ and pairwise mask-Hamming overlap is 0.631, while logit CKA passes. Mean-shift clustering in raw parameter PCA collapses posterior samples to one representative mode; the raw-parameter basin entropy is 0.0 nats with effective cluster count 1.0. The full-data CIFAR-10 ResNet-20 strengthens this separation: posterior samples can keep logit/final-hidden activation CKA above 0.93/0.91 while failing layer-sparsity, Hamming, or posterior-basin-count criteria.

Channel alignment does not change this conclusion: activation-channel and weight-correlation alignment both still fail layer KS and Hamming while retaining high activation CKA; a block-coordinate ResNet channel audit leaves global-channel posterior/ticket Hamming near 0.21 rather than ticket-like agreement; and an exact stage-1 enumeration validates the local solver on a small ResNet subgraph while exhaustive full-data graph isomorphism remains infeasible. A permutation-*invariant* metric closes this objection without enumerating permutations: per-tensor entropic Gromov–Wasserstein distances over channel graphs are relabeling-invariant by construction, and independent per-tensor couplings are a relaxation strictly more favorable to the posterior than any valid network symmetry. On the saved full-data SGLD artifact the metric is validated (a genuine producer/consumer channel permutation stays at the self-distance floor despite large coordinate Hamming distance), yet posterior samples stay at cross-seed ticket-diversity distance from their own-seed ticket in 5/5 seeds (0.96 – $1.06\times$ the cross-ticket mean): relabeling moves posterior masks no closer to the ticket at all.

The multi-chain and high-fidelity direct rows address the concern that a single local chain was too weak: a dense-start direct probe with 75 cyclical-SGLD posterior samples, and an independent-start version with 15 dense chain starts, both fail layer KS and Hamming with high CKA. A streamed exact joint-group direct probe over all 270,896 weight parameters closes the full-weight Gaussian gap: 25 joint-group posterior samples keep sample accuracy at 88.3%, posterior-to-chain-start Hamming is 0.0503, layer-sparsity KS is $p = 1.1\times 10^{-8}$, Hamming overlap is 0.000, logit/final-hidden activation CKA remains 0.937/0.920, and all 25 joint-group samples still collapse to one parameter-PCA basin.

Multimodal approximations: deep ensembles and tempered chains. Every approximation above is a local or single-chain family, so we add the two standard multimodal constructions. A *deep ensemble* of ten independently initialized full trainings—the approximation Wilson and Izmailov [20] advocate—keeps accuracy (0.880 versus dense 0.882) yet fails layer-sparsity KS ($p = 5.3 \times 10^{-7}$) and Hamming overlap (0.000) while passing logit CKA (0.931), members below IMP 5/5; *parallel-tempered SGLD* (1–8 ladder, 4–18% swap acceptance) fails identically (KS $p = 4.7 \times 10^{-9}$, overlap 0.000, CKA 0.935). Multimodality alone does not move posterior supports toward IMP tickets.

Exploratory robustness: non-residual architecture (seed-level cell). A reviewer could attribute the failure to residual architecture. We treat this cell as an *exploratory robustness cell* rather than a confirmatory row: it evaluates on the test split directly and bounds the architecture axis only; the dataset axis is handled by the validation-selected CIFAR-100 locked test below. We run the seed-level part of the matched probe on a non-residual three-convolution TinyCNN (no skip connections, no global average pooling; we retain BatchNorm so this is a non-residual—but not BN-free—robustness point) at the same five-round $p=0.30$ schedule and 0.832 sparsity, reporting the per-seed direction checks and posterior-to-chain-start Hamming (the activation-CKA stage stalls on TinyCNN’s wide 4096-dimensional pre-classifier feature, so the joint-distribution gate axes were not computed). The seed-level direction is unchanged: IMP beats the dense chain start in 5/5 seeds (0.791 versus 0.781); SGLD posterior samples stay below both in 5/5 seeds (0.738); and posterior TopK masks track the chain start with posterior-to-chain-start Hamming 0.061 (95% CI [0.060, 0.062]); the joint-distribution axes for a non-residual architecture remain an open robustness item.

Dataset robustness: CIFAR-100 with a validation-selected locked test. The dataset axis is tested at two levels on CIFAR-100 ResNet-20-w16 at the same $p=0.30$ schedule and 0.832 sparsity. First, an exploratory five-seed direct probe on the full training set (test-split evaluation): the seed-level direction holds (IMP beats dense 5/5 at 0.636 versus 0.629, posterior samples below both 5/5 at 0.617); on this harder task IMP supports coincide with dense magnitude (chain-versus-tickets Hamming overlap 1.000), so posterior masks clear the Hamming-overlap axis (0.816) while still failing layer-sparsity KS ($p = 1.3 \times 10^{-6}$) and activation CKA—4/6, as the rank-bounded intuition (§3) predicts. Second, the CIFAR-10 validation/locked-test protocol: a held-out 10% validation split selected the row and a single locked final-test rerun reproduced it exactly (layer KS $p = 9.0 \times 10^{-8}$, Hamming overlap 0.000, logit CKA 0.884, 3/6; IMP beats dense 5/5 at 0.620 versus 0.596, posterior 5/5 below both at 0.584—a 3.6-point gap to IMP). Under proper split separation IMP no longer collapses onto dense magnitude and the Hamming axis fails again, so the locked CIFAR-100 result is *stronger* than the exploratory probe suggested.

Support-equivalence is graded, not binary. The criterion behaves as a graded property in covariance fidelity rather than a single pass/fail switch. Of every posterior approximation tested, exactly one—the rank-128 Hessian-plus-diagonal Laplace direct probe—crosses any gate axis: its 50 posterior samples pass the Hamming-overlap threshold (0.816) with high logit/final-hidden activation CKA (0.932/0.910), but still fail the layer-sparsity KS threshold ($p = 2.0 \times 10^{-6}$) and collapse to one parameter-PCA basin. The pass is replicable, not single-run noise: an independent five-seed group (seeds 5–9) under the identical protocol reproduces the full profile (Hamming overlap 0.749, layer KS $p = 1.5 \times 10^{-6}$, one basin, same 5/6 axes). Neighbouring fidelities bracket it: rank-32/64 movement rows do not cross the mask-overlap threshold and the exact 270,896-weight joint-group row returns to Hamming overlap 0.000. A posterior account therefore has reproducible traction in exactly one narrow band—one family, one rank, one of three gate axes—reported as a partial-equivalence regime, not as support for the hypothesis: a replicated single-axis pass is still not the distributional mask/basin equivalence the hypothesis requires.

The layer-sparsity KS values in these direct rows are descriptive diagnostics over correlated posterior samples, not independent sample-level significance tests. The saved full-data SGLD artifact permits a seed-level paired reconstruction: posterior samples are farther from the same-seed IMP ticket than the chain-start support in 5/5 raw seeds and 5/5 activation-aligned seeds, and an independent seeds 5–9 extension run with saved masks reproduces the direction in 5/5 further seeds—10/10 across both groups, two-sided sign $p = 0.002$. The cyclical-SGLD multichain, rank-128 low-rank Laplace, and joint-group Laplace direct rows also write per-seed `mask_artifacts.npz` for the same on-demand paired audit; pooled-sample KS p-values for them remain descriptive diagnostics.

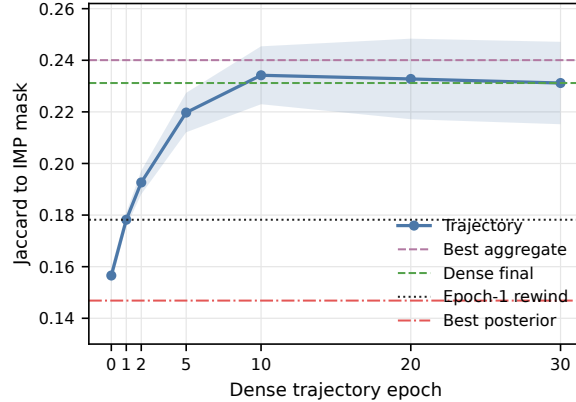


Figure 3: CIFAR-10 ResNet-20 dense-trajectory controls. Magnitude supports along the matched dense trajectory, especially around epochs 10–30, and the best aggregate trajectory magnitude score are closer to the IMP mask than any global posterior support in the movement diagnostics.

5 What Winning Tickets Are: A Trajectory and Process Account

The posterior-mode account fails the gates, but a winning ticket still demands a positive explanation. We give one: a winning ticket is a *trajectory and pruning-process object*, with a trajectory-magnitude component that any dense run exposes and a process-selected IMP-only residual that only iterative pruning produces. Neither part is recovered by posterior structure.

Dense trajectory magnitude is the first part, and it is useful: in the matched CIFAR dense-trajectory probe, trajectory magnitude supports are closer to IMP than posterior supports, and fixed-mask training shows that final dense, epoch-10, and RMS trajectory masks are much better than random. They are still below IMP, so the remaining question is what IMP adds.

Residual swap and removal-order controls localize the gap to a process-selected IMP-only residual. Anatomy, predictor-mask, cross-seed, direct-transfer, base-compatibility, posterior-ordering, and stratified controls give the same localization. Learned mask baselines do not replace that residual: Gem-Miner-style, Bernoulli/Concrete variational pruning, and the hard-concrete L0 gate baseline are below IMP in the current CIFAR support/accuracy rows and do not rescue the calibration/OOD tradeoff. Their learned objective can change uncertainty behavior, but not into a ticket-support explanation.

The strongest process controls test whether simpler score composition explains the residual. A tensor-matched excluded oracle stays below the round-selected IMP residual, and a tensor+score-matched oracle remains below it even after matching parameter tensor and within-tensor round-score deciles. Three residualization checks then confirm the residual is not a posterior artifact. A residualized round-score projection that removes base, dense-final, and final-IMP magnitude subspaces reduces both accuracy and oracle overlap; removing the diagonal-Laplace posterior score subspace (posterior RMS, standard deviation, and RMS-minus-dense features) leaves a 0.23-point round-vs-residualized accuracy gap (5/5 positive seeds, 95% CI $[-0.03, 0.48]$) with oracle overlap dropping 0.1923; and removing a learned trajectory/process subspace leaves a 0.48-point gap (5/5 seeds, 95% CI $[0.29, 0.67]$) with overlap dropping 0.1890. The learned process subspace captures broad directions but does not substitute for the particular residual coordinates IMP selects.

Positive prediction: cross-seed mask transfer. The trajectory and pruning-process account makes two falsifiable predictions that a posterior-mode account (which predicts seed-invariant support, hence little within/cross gap) does not: (P+1) same-seed IMP masks beat masks transferred from another seed, because the trajectory selects support per seed; (P+2) cross-seed masks still carry above-random support precision, because the trajectory-magnitude component is partially shared. Both hold in the five-seed transfer probe: same-seed accuracy minus IMP is -0.0090 ($[-0.0111, -0.0068]$) versus cross-seed -0.0235 ($[-0.0269, -0.0201]$), a 1.5-point drop (P+1); and cross-seed added support precision is 0.2388

([0.2334, 0.2442]), roughly twice the random baseline 0.1264 (P+2).

6 Discussion

These results constrain a simple Bayesian interpretation of lottery tickets. The posterior signal exists in a weak sense—posterior-induced masks are not random—but the same signal is already present in the dense, rewind, trajectory, or chain-start magnitude support. The explicit distribution-equivalence audit turns this into a negative result rather than a single-metric impression, and the evidence favors a trajectory/magnitude subspace account plus a process-selected residual over a posterior-mode account.

This adjudicates the two competitors of Section 2 in favor of the axial-subspace reading. Paul et al. [15] hold that IMP masks encode trajectory axial subspaces, not posterior structure; the posterior-mode reading predicts the opposite. Our gates test that prediction and it fails: trajectory magnitude and the IMP residual carry the ticket signal, posterior supports do not, and the graded rank-128 exception passes only one axis at one fidelity. The evidence is consistent with the axial-subspace account and inconsistent with a posterior-mode account in the tested settings.

This conclusion is compatible with mode-connectivity results, and a generated linear connectivity audit makes the point concrete: MNIST and Fashion-MNIST dense-to-IMP barriers are nearly zero (0.0026 and 0.0395) while posterior support stays below the chain-start control, and CIFAR-10 long SGLD and SWAG barriers are large (3.0827 and 3.7402) while posterior support stays tied to chain-start support. Linear barriers are orthogonal landscape diagnostics, not evidence of posterior-ticket equivalence.

Scope of the claim. The claim is not that Bayesian neural-network posteriors are useless, nor that no posterior approximation could ever correlate with IMP coordinates. The confirmatory evidence in this paper is the five-seed MNIST/Fashion-MNIST sparsity sweeps, the CIFAR-10 ResNet-20 epoch-1-rewind direct probe across fourteen posterior approximation configurations, and the validation-selected CIFAR-100 ResNet-20-w16 locked-test row; the non-residual TinyCNN seed-level cell is an additional robustness check under the same matched-sparsity schedule. The claim is narrower: under the posterior families and CIFAR-10/MNIST/Fashion-MNIST settings tested here, posterior-induced sparse supports do not satisfy the support-equivalence claim once dense, rewind, trajectory, and pruning-process controls are included; useful ticket structure is better explained by trajectory magnitude plus a process-selected IMP residual.

7 Limitations and Next Experiments

The evidence is broad, but the paper should still be read as a controlled negative result under the posterior approximations we can currently test, not as a theorem about all Bayesian neural-network posteriors. We also note an explicit multiple-comparisons exposure: confirmatory rows pool five seeds against four to six thresholds across fourteen posterior configurations, so individual pooled p-values are descriptive diagnostics. A generated family-wise audit quantifies the exposure over the sixteen distinct full-data direct configurations ($16 \times 6 = 96$ trials): all sixteen layer-sparsity KS rejections survive Holm–Bonferroni at $\alpha = 0.05$; the two single-axis Hamming-overlap passes (rank-128 and CIFAR-100) match the expected single-axis pass count under pooled per-axis base rates; and no configuration passes the full joint gate. A TOST reanalysis at the paper’s own materiality margin ($\delta = 0.005$ Jaccard) equivalence-passes all eight Gate1 tracking rows, with all 35 CIFAR movement rows one-sided non-superior at the same margin; the four independent seed-level direction checks (saved-artifact SGLD seeds 0–4 and 5–9, TinyCNN, CIFAR-100) jointly have global-null probability 9.5×10^{-7} , and the headline SGLD row alone spans ten seeds (10/10, two-sided sign $p = 0.002$). The conclusion thus rests on family-wise-corrected rejections, equivalence-tested tracking, and joint seed-level direction, not on any single pooled p-value. The main remaining limitations are:

- alignment robustness. Activation-channel and weight-correlation alignment, record-level post-hoc matching, the block-coordinate global channel audit, and exact stage-1 enumeration are negative or bounded; exhaustive full-data graph isomorphism remains infeasible and unimplemented;
- validation/test separation. The selected CIFAR-10 SGLD direct row was rerun on a held-out 10% validation split and a locked final-test rerun has been completed (posterior-sample accuracy 0.872, 4/6

thresholds, matching the validation row); the CIFAR-100 row follows the same protocol (validation-selected, locked test reproducing it at 3/6);

- BatchNorm posterior policy. ResNet posterior samplers include BatchNorm buffers in sampled state dictionaries under the default train-buffer policy; the full CIFAR SGLD/cSGLD sweep over freeze, recalibrate, and dense-buffer policies keeps the negative direction in 5/6 rows, with frozen-BN vanilla SGLD reported as an invalid sampler setting;
- direct distribution statistical unit. Pooled direct mode/ticket p-values over posterior samples are descriptive over correlated samples; the SGLD, cSGLD multichain, rank-128 Laplace, and joint-group Laplace direct rows all write per-seed `mask_artifacts.npz` for seed-level paired audits, and the headline SGLD direction now spans ten seeds (sign $p = 0.002$);
- full-network exact/full-covariance Laplace or another still broader high-fidelity posterior baseline: the covariance ladder up to all-weight joint groups, 20-snapshot SWAG, and subspace HMC is covered, but the exact dense full-network CIFAR posterior remains absent;
- broader learned-mask distribution variants. Gem-Miner-style, variational, and hard-concrete L0 gates do not rescue support, accuracy, calibration, or CIFAR-100 OOD behavior; additional permutation-aware learned-mask distributions would make this negative claim cleaner;
- additional process-causality controls. The residual-swap, anatomy, predictor-mask, transfer, ordering, stratification, IMP-process, tensor+score-matched round-exclusion, residualized-score projection, posterior-residualized, and learned-subspace residualized controls do not exhaust the possible causal interventions on the pruning process;
- Reproducibility Statement. The repository ships an environment lock, Makefile gates, artifact verifier, CI workflow, release archive, source snapshot, verification containers, and an external-validation run-book; public archive, public repository, and external CI receipts are recorded against the current archive SHA and source commit, and the one remaining external receipt is an independent GPU-container run.

Since the aligned, multi-chain, Laplace-family, locked final-test, and BatchNorm-policy CIFAR runs preserve the current pattern, the next decisive empirical gap is an even stronger high-fidelity posterior baseline; if it stays near the chain-start and rewind controls, the negative result is the right paper, otherwise the claim should weaken to a partial-equivalence statement (the rank-128 row passes 5/6 thresholds and is the strongest example in the current evidence).

Ethics Statement

This work uses standard public benchmark datasets and does not introduce human subjects data, private personal data, surveillance data, or safety-critical deployment claims. The main ethical risk is scientific overclaiming: reading the negative result as a theorem rather than a controlled empirical result under the tested architectures, approximations, sparsities, and budget. We mitigate it by stating the operational criterion, reporting negative and bounded findings, preserving open limitations, and packaging code and artifacts for reproduction. Final submission still requires the human authors to review and acknowledge the applicable ICLR Code of Ethics.

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